



Munhall-based business focuses on affordable, fossil-free energy

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By Annie Siebert / Pittsburgh Post-Gazette

After 15 years working as an engineer for various corporations -- including steel, plastics and pharmaceutical companies -- Steven Kovacik decided to take his knowledge of "virtually every aspect of manufacturing" and start his own business.

He's focusing his expertise on creating an inexpensive energy generator that doesn't use fossil fuels.

"I've always thought that our future as a species on this planet was contingent on finding sustainable forms of energy," Mr. Kovacik said of starting ZeroFossil, based in Munhall.

ZeroFossil products launched in March, but prior to that, Mr. Kovacik spent two years developing a sustainably powered generator called the "JuiceBox," a small, portable generator that can run on solar, wind, hydroelectric or human power via a "Bikerator," which is exactly what it sounds like -- a small device that can attach to any adult-size bicycle to create human-powered energy.

The Bikerator was designed to create enough human-powered energy to run a flatscreen television and a video game system. Mr. Kovacik said parents would use it to let kids power their video games -- as long as the child can sustain the energy needed to power the game, they could play.

"What happens is, kids will do that for a couple days, couple weeks, then they'll get bored and go outside, which is what the parents wanted in the first place," he said with a laugh.

Mr. Kovacik said his experience with industrial manufacturing gave him a strong foundation for how to make the products, while his background in chemical engineering gave him the knowledge to make things that didn't exist before.

Another product, ZeroFossil's Integrator, was designed to collect energy from solar panels (or other sustainable sources) and convert and store the energy for use in a home. It costs about \$5,000, plus installation. Mr. Kovacik said a traditional solar panel energy system in a home could cost between \$20,000 and \$40,000, making the Integrator more cost-effective for people who want to install a sustainable source of energy in their home for emergencies.

"If the power were to go out for two to three weeks, they could sustainably stay powered -- keep your food cold, keep your lights on, enough to stay comfortable -- indefinitely," Mr. Kovacik said.

During nonemergency scenarios, using the Integrator can help reduce homeowners' electric costs.

"While the grid is working properly, the Integrator converts all excess electricity to grid-compatible [energy] and sends this back into your house to reduce your dependency on the grid, hence reducing your electric bill," Mr. Kovacik said in an email. "Think of a backup generator that pays for itself every time the sun rises or the wind blows."

One of ZeroFossil's latest projects was creating an entirely off-grid energy system at the site of the Carrie Furnace in Rankin, which was once the iron-producing facility for U.S. Steel Corp.'s Homestead Works. It has been shut down since 1986, and the Rivers of Steel National Heritage Area and its volunteers are working to restore the site into a museum.

The entire Carrie Furnace site is powered on three 200-watt solar panels, a total of 600 watts of generation for the whole site. Bomb shelter lights and an old control panel in the former "stock house" are lit, and 1-watt LED lights illuminate areas formerly shrouded in shadows. Beams of sunlight stream through rusted holes and broken walls, and it's difficult to tell the LED lights from bits of sunlight.

Rivers of Steel National Heritage Area initially wanted only the original light fixtures to be lit, but Mr. Kovacik said he wanted to go further, lighting control panels and warning lights throughout the site.

The original fixtures in the stock house held 120-volt incandescent bulbs, and now they have 12-volt compact fluorescent bulbs. All of the lighting at the furnace comes in at just under 150 watts, which used to be the wattage of a single light in the plant.

"This place used enormous amounts of energy when it was operational," Mr. Kovacik said.

Now, solar power is lighting and powering the whole plant on minimal energy, "all as a way of showing future generations where previous generations came from," Mr. Kovacik said.

Sustainability technician Mark Rawlings was tasked with running about 2,000 feet of wire throughout the furnace after Mr. Kovacik broke his foot right as ZeroFossil was scheduled to begin work on the furnace. Last week, Mr. Rawlings, a rock climber, moved swiftly and skillfully among the rusted pipes of the furnace, neatly zip-tying wires to pipes, sometimes in the dark.

He shuffled along metal pipes about 6 inches in diameter and 2 feet apart.

"This place was made for climbing," he said. "The great thing about wiring things here is that there are pipes everywhere. I just climb around and put zip ties up."

By the end of this week, Mr. Kovacik and Mr. Rawlings will complete work at the furnace. Lights, security cameras, a small theater and a flat-screen television at the site will all be completely powered by the sun.

Mr. Kovacik practices what he preaches -- he said 95 percent of the materials for ZeroFossil products are American-made, and the solar panels and all of ZeroFossil's products are manufactured locally. Additionally, no fossil fuels are used in production -- everything is manipulated, fabricated, assembled and installed without the use of fossil fuels.

ZeroFossil bartered its services with Braddock-based Fossil Free Fuel to convert a delivery vehicle to run on veggie oil.

"Fossil fuels are not the best form of energy for our planet" because they're finite, Mr. Kovacik said. "If the human race wants to continue to inhabit this planet, we have to wean ourselves off of fossil fuels at some point."

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